



Ionel Aungurencei

Nationality: Romanian | **Phone number:** (+40) 756271599 (Mobile) |

Email address: ionel.aungurencei@gmail.com | **LinkedIn:**

<https://www.linkedin.com/in/ionel-aungurencei-00a64b62/> |

Address: Iași, Romania (Home)

● ABOUT ME

I am a fast-learner and technology fan. I like to see how things work and i always want to improve them.

● WORK EXPERIENCE

06/2023 – CURRENT Iași, Romania

EMBEDDED SYSTEMS SOFTWARE DEVELOPER KATTY FASHION SRL

Spearheading initiatives to enhance digitalization in the fashion industry through advanced manufacturing devices.

Contributing to the improvement of manufacturing processes and systems.

Working occasionally on cutting edge EU-funded projects where I help integrating the technology provided into the Fashion Factory of the Future and I contribute on designing the best tools for the challenges proposed.

Limited time job, more like a hobby, a way to learn new technologies and make an impact in fashion industry.

Keywords: IIoT, CPPS, Product/Process Digital Twin, AI

07/2021 – 12/2023 Iași, Romania

AEROSPACE EMBEDDED SYSTEMS SOFTWARE ENGINEER ENEA SERVICES ROMANIA SRL/AROBS DEVELOPMENT AND ENGINEERING SRL

Team and technical leader for the Software discipline of one module in a State-of-the-Art autonomous satellite with both military and civil applications.

The last 6 months took more developer tasks to be able to finish the needed functionalities in time and in budget.

As technical leader, I was in charge of writing documentation for the project according to ECSS standards and taking all the needed decisions based on the requirements of the client. I was also in charge of all the technical communication with the client.

The project was developed in C, had a dualcore ARM A9 processor with FPGA area that was used by a set of algorithms.

FreeRTOS was used in AMP configuration, with custom inter-processor communication.

I worked also with an FPGA engineer to receive the required IPs with the needed configuration.

I developed a file system interface between the PUS services and elm-fatfs.

Ported openCV to baremetal compilation without POSIX interface to be usable with FreeRTOS and the algorithm library that the client provided.

I worked with the QA responsible to provide the necessary information for the quality matrix and for risk anagement.

Keywords: C, openCV, FPGA, FreeRTOS, AMP, ECSS, file system.

11/2018 – 07/2021 Iași, Romania

EMBEDDED SYSTEMS SOFTWARE DEVELOPER PREH ROMANIA SRL

Software Engineer in E-Mobility Department.

I was nvolved in the pre-development for a DCDC converter.

I implemented SW functionalities for Aurix. I configured the GTM peripheral to generate the needed waveform based on the control strategy. I worked on the control algorithm and external ADC configuration. I helped HW team reduce costs by changing some integrated chips and moving to SW the filtering of signals which actually allowed to use better control algorithm.

I worked with the the BMS team. I was involved updating the platform project to fit new client requirements with tasks in requirement analysis, adding new functionality, bugfixing and extensive testing of safety relevant code. I worked on the internal power-saving features (entering/exiting sleep, RTC), updating SW requirements, testing the whole system, blackbox unit testing of Safety Controller code, design and unit test for ASIL code. The team was impressed about my findings resulted from unit tests, how I created SW design and unit test requirements. I found a bug in the existing code which was really hard to explain.

In the Emobility team, I worked on a complex project unifying multiple devices into 1. I worked on the SW for the part that assures high power transfer.

I developed features requested by client and working close with HW so we deliver a stable project.

I proactively contributed to the project with suggestions and optimisations for other disciplines (HW, System).

I used PLECS to do SW-in-the-loop to be able to simulate the HW control algorithm that used PID and 2p2z control loops before the HW was available.

I was in charge with Software architecture for a subsystem of 3 controllers that communicated with a main controller.

I was also a technical lead for SW implementation team of 5, for a SW having to control complex HW circuits and having the HW expertise to propose improvements or to review changes during later phases of development.

I created the SW architecture for each microcontroller (16bit dsPIC), designed the Bootloader, the synchronization between bootloader and application SW and took in account the available peripherals and the constrained resources so that we could do the functions needed.

I was in close contact with the HW team so that they would design an cost effective HW based on SW capabilities.

I found many issues in the HW schematics that were overlooked and suggested an alternative for safety related functionalities, one with the most impact being ADC plausibilisation which removed more than 30 components from the PCB.

I participated in FMEDA meetings and worked on the SW Safety Concept.

Keywords: DCDC converter, Aurix, dsPIC, Control strategies, BMS team, Requirement analysis, Bugfixing, Safety relevant code, Power-saving features, RTC, ASIL code, Battery cell monitoring, PLECS, SW-in-the-loop, PID, 2p2z, Software architecture, Bootloader, Synchronization, Peripherals, Constrained resources, ADC plausibilization, FMEDA, SW Safety Concept.

01/2018 – 11/2018 Iasi, Romania

EMBEDDED ENGINEER CODEFICUM 42 SRL

I did various projects and consultancy work.

Most relevant:

- single chip data capture device with 64 ADC channels (PCB + soldering + demo code)
- arcade box with buttons and touch screen with internet connection to Raspberry Pi server that took data From Facebook API and had Bootstrap front end and Python/Flask backend (Mechanical design of enclosure, PCB, firmware in C with graphical library for touchscreen, front/back end coding, assembling)
- Raspberry Pi BLE gateway (python + bluez stack)
- front panel for laser device with self-capacitance buttons and LEDs (PCB, assembling, firmware with bootloader)
- vehicle 360degree system (MVP setup consultancy)
- location-aware BLE 5.1 devices

Keywords: data acquisition, PCB assembly, custom embedded devices, RaspberryPI, BLE. Direction Finding

01/2015 – 12/2017 Iasi, Romania

SOFTWARE ENGINEER INATECH SRL

Worked as remote extern employee for german automotive OEM within the low-level communication driver team.

My team was in charge of developing the Can, Lin, Flexray and Ethernet drivers for AUTOSAR , adding features, making quality assurance artefacts, developing tests (unit, component, integration) and making bug fixes. I was also assigned various tasks to help other colleagues.

Main work packages I worked on:

- Can Driver development support (adding new small features, fixing problems, support for integrators and customers)
- IPv6 development (adding IPv6 components for the Ethernet Stack, making unit and component tests)
- ComStack gateway (added gateway configuration and tests)

Keywords: AUTOSAR 4.x, CAN, LIN, FLEXRAY, ETHERNET, IPv6, CANoe, OAW code generators, Cantata Unit testing, Python, Makefile, Jenkins CI

● EDUCATION AND TRAINING

2010 – 2014

COMPUTER ENGINEERING Gheorghe Asachi Technical University of Iași

During studies, I worked with a professor on lots of embedded software/hardware contests, where I learned how to actually apply the knowledge received.

Thesis Field-oriented control of PMSM

● LANGUAGE SKILLS

Mother tongue(s): **ROMÂNĂ**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLEZĂ	C2	C2	C1	C1	C1
FRENCH	A2	A2	A2	A2	A1
ITALIAN	B1	B1	A2	A2	A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● ADDITIONAL INFORMATION

DRIVING LICENCE

Driving Licence: B

PROJECTS

Projects - jan 2013 - LCD GUI for a project that measured liquid's viscosity
 - jun 2013 - nov 2014 - developed devices and sensors for smart fish aquarium (water quality sensors, battery management and backup)
 - jul-oct 2014 developed GSM-enabled device to count products on conveyor belt
 - oct 2014 - 32bit 1MHz logic analyzer
 - dec 2014 - feb 2015 - hardware parts for the first Escape Room in Iași
 - mar - may 2015 - barcode generator
 - oct 2017 - 10Mbps custom serial interface
 - mar - may 2018 - PetFeeder - design PCB and firmware for WiFi enabled device to feed home pets while away
 - may 2019 - september 2020 - BLE 5.1 direction finding project - contributed to RF antenna design and some HW layout. Did initial programming of nRF52811 and run some tests.

JOB-RELATED SKILLS

Job-related skills

I worked with a wide variety of microcontrollers (ATmega16, Renesas RX/RH, Microchip PIC16/dsPIC33C, Freescale/NXP Kinetis(Cortex M7 (NXP KV58F1M0), M0, M3, M4), STM32 (STM32F429, STM32F769 - both with LCD display with multi-touch interaction), Nordic Semiconductor (nRF52), BeagleBone Black + PRU/

PRUSS/PRU-ICSS, LittleGL graphics library, I can do code optimizations, AUTOSAR/MISRA compliant code, assembler inserts, RTOS integration, peripheral HAL drivers.

I program mainly in C with some Assembly, I know more than enough (for an embedded dev) of C#, Python, Javascript. I can read, understand and do small changes for many more programming languages.

I have made different PCBs (PCB design, PCB BOM optimization, PCB assembly) using Kinetis/STM32/ Nordic controllers in Altium Designer, Altium Circuit Maker and KiCAD, from idea to BOM and assembly to functional device.

CERTIFICATIONS

Certifications

- apr 2019: Attended training "Functional Safety acc. to ISO26262 with ISO2511" from Exida.com Gmbh.

STUDENT CONTESTS AND PROJECTS

Student contests and projects

- feb-apr 2012 participated at ElectroMobility contest organized by Continental where I was in charge of doing the PCB design of a BLDC motor controller

- apr-oct 2012 - teacher at "How to think in C" labs

- summer 2012 - I did various project with ATmega16 as summer practice

- oct 2012-mar 2013 - I participated to "Freescale Cup" competition where my team won local contest and participated in Paris and the european stage. The task was to control a toy car to follow a black line on white surface. I did the PCB and parts of the control algorithm.

- nov 2012 - jul 2014 - worked on my thesis, about a PMSM motor controller hardware. I did sample PCBs and implemented algorithm.

- nov 2012 - second place BEST Engage IT by doing an ASP.net website from scratch

- nov 2012 - may 2013 - participated at OpenDoors Sibiu with an autonomous robot that had to pop balloons of certain color.

- nov 2012 - may 2013 - participated at EletroMobility 2 with an autonomous maze navigating robot with remote control wireless GUI and I was in charge of doing the obstacle avoidance software

- mai 2013 - Hard and Soft Suceava contest - we had to control a toy helicopter and to detect some geometric forms

- nov 2013 - apr 2014 - "Freescale Cup" competition where my team qualified at the national contest to participate at the European stage in Erlangen at Fraunhofer IIS. The task was to control a toy car to follow a black line on white surface. I changed the controller from last year and we also changed the control algorithm.

- apr-may 2014 - help the Embedded System training for AIESEC Iași.

- may 2014 - Hard and Soft Suceava contest - we had to make a robot to explore foreign planets' terrain and gather weather data.

- may 2014 - won ITMarathon Embedded section. The theme was doing an efficient control of LEDs and a receiver for IR remote controller on Infineon platform